Decision Model Training Flow

Input utterance x

En | Bn | CM | Rom

↓

Decision prompt

↓

Tokenizer + embedding

text → token vectors

↓

Pre-trained LLM M_0

Qwen3.5-0.8B

Transformer decoder × N

+ LoRA adapters (r = 16)

↓

Intent scoring

decision head: p(c | x, C)

alarm_set	▮▮▮▮▮▮▮▮	**0.84**
alarm_query	▮	0.09

↓

Predicted intent ŷ

one of MASSIVE's 60 intents

↓

Supervised loss

compare ŷ with gold intent *y*

update LoRA + head weights (backprop)

after training

M_{dec}

Decision model